Younghoo Henry Cho

Engineer
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Summary

Engineer on the connected-car services (CCS) platform for Hyundai, Kia, and Genesis across North America, managing, developing, operating, and analyzing the full data flow: Kubernetes-based services, Java/Spring and Python data platforms, and carrier API integrations. Ph.D. (2026) in verifiable data sharing: built a blockchain platform that lets organizations verify where shared data came from and how it changed. Seven years across backend systems, data infrastructure, and applied machine learning; two patents (one granted, one pending) and peer-reviewed publications.

Experience

System Engineer, Solomon America (Duluth, GA) **Sep 2026 – Present**

System Engineer, Hyundai AutoEver America

- Manage, develop, operate, and analyze the CCS platform that connects Hyundai, Kia, and Genesis vehicles in the U.S., Canada, and Mexico to each brand's services: vehicle factory, carrier network, vehicle-control, and remote-command data, from ingestion to application services, on Kubernetes.
- Diagnose production incidents to root cause in the platform's Java/Spring services and implement the fixes; review the code for security weaknesses and ship hardening changes.
- Build and operate carrier integrations (Verizon, Bell) that provision and bill vehicle connectivity: API and callback flows for line activation and deactivation, data-plan changes, and usage and billing events.
- Develop and operate five internal data platforms (Python/Django) and 10+ automation tools for monitoring, analyzing, correcting, and reporting CCS data across North America.
- Define integration designs and scope estimates for new in-vehicle data products.

Graduate Research Assistant (Data Infrastructure), University of Florida **Aug 2022 – Aug 2026**

- Designed and built DEMETER, a blockchain system that lets an organization confirm where shared data came from and how it was changed, without having to trust or contact the organizations that handled it earlier.
- Built the core in Go on Hyperledger Fabric with a REST API, where each organization records a file's identity and handoffs under its own credentials with no central operator, and ran the multi-organization network on Docker with CI-driven build and test.
- Verified the system on soil-carbon data: it catches altered files, blocks records that cite unregistered sources, and tells apart two records that share a sample ID but hold different data.
- Added an LLM assistant so non-technical users can query the ledger in plain language, behind a write-confirmation gate: the assistant reads freely but cannot change ledger state without explicit approval.

AI Engineer, Billion21 **Dec 2019 – Aug 2022**

- Built computer-vision and machine-learning models in Python that read fish feeding activity from camera and sensor input to set feed amount and timing per species in an automated feeding system.
- Co-invented a computer-vision method for seaweed spore selection that replaced manual microscope inspection (granted patent).
- Co-invented a blockchain system that records each handoff of seaweed from farm to buyer so any party can verify its path (patent pending).

AI Consultant, Global BioAg Linkages **Jan 2020 – Dec 2020**

- Planned AI/ML adoption for agriculture and aquaculture companies, assessing technical feasibility and market viability for each.

Education

University of Florida, Ph.D., Agricultural and Biological Engineering, GPA: 3.95/4.0 **Aug 2022 – Aug 2026**
Texas Tech University, B.S., Computer Science, Minor in Mathematics, GPA: 3.8/4.0 **Aug 2015 – Aug 2022**

Selected Publications

10 total, including under review

- Cho, Y., Yu, Z., Ampatzidis, Y., Nam, J.-S. *Facilitating a Future Agricultural Data Ecosystem: A Cross-Sectoral Review of Blockchain Application in Data Sharing*. Precision Agriculture, 27, 138 (2026).
- Cho, Y. et al. *DEMETER: An Integrated Trust Infrastructure for Shared Agricultural Data*. Under review.
- Cho, Y. et al. *An Enabling Infrastructure Layer for Origin Verification and Transfer Tracing of Shared Agricultural Data Files*. Under review.
- Cho, Y. et al. *An Infrastructure Layer for Tracing Transformed Agricultural Data Back to Its Sources*. Under review.

Patents

- *Blockchain-Based Seaweed Farming Data Management and Smart Supply Chain Tracking System*. KR10-2024-0011399, pending.
- *Methods for Selecting Spores from Laver Farm Devices*. KR102034354B1, 2019 (granted).

Skills

Languages: Python, Java, Go, SQL, JavaScript, C, C++, R
Data Engineering: Apache Airflow, Apache Spark, Apache Kafka
Databases: PostgreSQL, MongoDB
Distributed Systems & Backend: Django (Python), Spring (Java), Node.js, Express, REST APIs, gRPC, Hyperledger Fabric
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux
ML & AI: PyTorch, TensorFlow, OpenCV, YOLO, SAM, XGBoost, LightGBM, LSTM, Phi-3, LangChain, Ollama